Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2004-II-Q02] | Part A (Q1-Q5)

A jar contains ten red and ten blue candies. One person draws two uniformly without replacement, then another draws two from the remainder. The probability of equal unordered color combinations is m/n reduced. Find $m+n$.

罐中十红十蓝糖，第一人等概率不放回取两颗，第二人从余下再取两颗。两人所取无序颜色组合相同的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2006-II-Q05] | Part A (Q1-Q5)

Two independent identical biased dice have opposite face labels summing to seven. Four faces each have probability $1/6$; face F has larger probability and its opposite smaller. The chance of total seven is $47/288$. Express $P(F) = m/n$ in lowest terms and find $m + n$.

两个独立同分布的偏骰，其对面点数和为七。四面概率均为 $1/6$，面 F 概率更大，其对面更小。总点数七的概率为 $47/288$。若 $P(F) = m/n$ 最简，求 $m + n$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-I-Q02] | Part A (Q1-Q5)

One urn holds four green and six blue balls; another holds sixteen green and N blue. Independent uniform draws give a same-color probability of 0.58. Find N.

第一罐四绿六蓝，第二罐十六绿及 N 蓝，各独立等可能抽一球，同色概率为 0.58，求 N。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-II-Q02] | Part A (Q1-Q5)

Three events A, B, C have each exactly-one region probability 0.1 and each exactly-two region probability 0.14. Given $\Pr(C \mid A \cap B) = 1/3$, reduce $\Pr(\overline{A \cup B \cup C} \mid \bar{A}) = p/q$ and find $p + q$.

三个事件的每个恰一事件区域概率为 0.1，每个恰两事件区域概率为 0.14。已知 $\Pr(C \mid A \cap B) = 1/3$，将无事件发生且已知 A 不发生时的条件概率约为 p/q，求 $p + q$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2015-I-Q05] | Part A (Q1-Q5)

Five sock colors have two socks each. On each of three successive days, two socks are sampled uniformly from those remaining, without returning any. If the probability that the third day is the first matching-color pair is reduced m/n, compute $m+n$.

五种颜色的袜子各两只。连续三天，每天从剩余袜子中等可能取两只，不放回。第三天首次取到同色一双的概率为最简分数 m/n。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2015-II-Q02] | Part A (Q1-Q5)

The four grade levels contain $40\%, 30\%, 20\%, 10\%$ of a school. Their Latin-enrollment rates are respectively $100\%, 80\%, 50\%, 20\%$. If the probability that a uniformly chosen Latin student is in the second grade level is reduced m/n, find $m+n$.

学校四个年级人数依次占 $40\%, 30\%, 20\%, 10\%$，各年级选修拉丁语的比例依次为 $100\%, 80\%, 50\%, 20\%$。从拉丁语学生中等可能选一人，其属于第二年级的概率为最简分数 m/n。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2016-I-Q02] | Part A (Q1-Q5)

Two independent six-sided dice have $\Pr(k)$ proportional to the face value k, for $k = 1, \ldots, 6$. If the probability of a total of 7 is reduced m/n, find $m + n$.

两个独立六面骰的各面概率与点数 k 成正比，$k = 1, \ldots, 6$。若总点数为 7 的概率为最简分数 m/n，求 $m + n$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2016-II-Q02] | Part A (Q1-Q5)

The probabilities of rain on Saturday and Sunday are $2/5$ and $3/10$. Sunday rain is twice as likely conditional on a rainy Saturday as conditional on a dry Saturday. If the probability of rain on at least one day is reduced a/b, find $a+b$.

周六、周日降雨概率分别为 $2/5, 3/10$。已知周六下雨时，周日下雨的条件概率是周六不下雨时的两倍。若至少一天降雨的概率为最简分数 a/b，求 $a+b$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2017-II-Q02] | Part A (Q1-Q5)

Teams T_1, T_4 contest one semifinal, while T_2, T_3 contest the other; winners meet in a final. In any match, T_i beats T_j with probability $i/(i+j)$, independently of other matches. If T_4 wins the title with reduced probability p/q, find $p+q$.

半决赛由 T_1 对 T_4、T_2 对 T_3，胜者进入决赛。任一场 T_i 对 T_j 的胜率为 $i/(i+j)$，各场独立。若 T_4 夺冠概率为最简分数 p/q，求 $p+q$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2021-I-Q01] | Part A (Q1-Q5)

Two runners contest six races. Runner Z wins the first; subsequently the winner of the previous race wins again with probability $2/3$. If the probability that Z wins exactly five races is m/n reduced, find $m+n$.

两人赛跑六次，Z 赢得第一次。以后每次前一场胜者再次获胜的概率为 $2/3$。若 Z 恰赢五次的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2022-II-Q02] | Part A (Q1-Q5)

Four players A, C, J, S receive uniformly random semifinal pairings; the winners meet in a final. Player A beats C with probability $2/3$. Either A or C beats either J or S with probability $3/4$. Match outcomes are independent. Express the probability that C is champion as reduced p/q and find $p+q$.

四位球员 A, C, J, S 随机等可能配对打半决赛，胜者争夺冠军。A 对 C 的胜率为 $2/3$；A, C 中任一人对 J, S 中任一人的胜率为 $3/4$。各场结果独立。若 C 夺冠概率为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2024-I-Q04] | Part A (Q1-Q5)

A lottery ticket chooses four distinct numbers from $1, \ldots, 10$. A random four-element winning set gives a prize for at least two matches, and the top prize for four matches. Given that a prize was won, write the probability of the top prize as m/n in lowest terms and find $m+n$.

彩票从 1 至 10 中选四个不同号码，开奖也等可能选出四个。至少匹配两个号码即获奖，四个全中为头奖。已知获奖，头奖的条件概率为最简分数 m/n，求 $m+n$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-II-Q06] | Part B (Q6-Q10)

Choose equally between a fair die and a die whose six-probability is $2/3$ and each other face has probability $1/15$. Roll the selected die three times. Given the first two are sixes, reduce the chance the third is also six to p/q and find $p+q$.

等可能选公平骰或一枚六点概率为 $2/3$、其余各点概率 $1/15$ 的骰子，固定该骰投三次。已知前两次均为六，第三次也为六的概率约为 p/q，求 $p+q$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2017-II-Q09] | Part B (Q6-Q10)

A deck has one card for each of the49 combinations of seven numbers and seven colors. Choose an eight-card set uniformly, conditional on covering all numbers and colors. The probability that one card can be discarded while retaining complete coverage is reduced p/q. Find $p+q$.

七个号码与七种颜色的49种组合各对应一张牌。等可能选择八张牌组成集合，条件是每种号码和颜色均出现。若可丢弃一张仍保持完整覆盖的条件概率为最简分数 p/q，求 $p+q$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2026-I-Q09] | Part B (Q6-Q10)

Roll a fair blank six-sided die six times. After roll j, place sticker j on the top face, covering earlier stickers there. Conditioned on stickers $2, 4, 6$ all being visible at the end, the probability of exactly one blank face is m/n in lowest terms. Find $m + n$.

公平空白六面骰掷六次，第 j 次后在朝上的面贴上编号 j，覆盖该面的旧贴纸。已知最终 $2, 4, 6$ 都可见，恰有一面空白的概率为最简分数 m/n，求 $m + n$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2010-II-Q13] | Part C (Q11-Q15)

Four players draw distinct cards numbered 1 through 52. The lower two form one team, the higher two another. Condition on two designated players holding $a, a+9$, and let $p(a)$ be their chance of being teammates. Find $m+n$ for the smallest value $p(a) \geq 1/2$, written as reduced m/n.

四人各抽取一张不同的 1 至 52 号牌，较小两牌为一队，较大两牌为另一队。已知指定两人持牌 $a, a+9$，其同队概率为 $p(a)$。将不小于二分之一的最小可能 $p(a)$ 约为 m/n，求 $m+n$。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2011-I-Q12] | Part C (Q11-Q15)

Six men and n women line up randomly. Given that each man has a male neighbor, let p be the probability of a run of at least four men. Find the smallest n for which $p \leq 1\%$.

六男与 n 女随机排成一行。已知每名男子旁边至少有一名男子，设至少四男连续的条件概率为 p。求使 $p \leq 1\%$ 的最小 n。

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